

## LAB ACTION PLAN FOR WEEK 10

### Working with minikube and Nagios

Hands-on practice of creating, running and scaling pods in minikube.

Running Nginx server on specified port number by explaining the Nginx monitoring tool

Running Nagios server and Understanding the Monitoring tool using Docker.

AWS-free Trier account Creation steps

Upload the screenshots for the tasks

#### Kubernetes:

Kubernetes is a tool that automates how we run and manage applications inside the container.

Dockers will only run containers, if in any case the container fails/stopped/killed, the docker will not help us, here is where Kubernetes plays an important role, Kubernetes cluster will be responsible in creating a new container and managing various containers.

**POD:** In Kubernetes, a Pod is the smallest deployable unit that you can create and manage.

#### Minikube:

Minikube creates a VM on your local machine and deploys a simple Kubernetes cluster with one node. It's a lightweight implementation. Minikube is a version of Kubernetes.

#### Nagios:

Nagios is an **open-source IT infrastructure monitoring tool**. It monitors

- Servers
- Network devices
- Applications and services

It **alerts administrators** when issues occur and notifies when they are resolved.

Step 1: Install Prerequisites

Before installing Minikube, ensure the following are installed:

#### **Virtualization Support:**

Verify virtualization is enabled:

## Hypervisor:

Minikube supports multiple hypervisors (e.g., **Hyper-V**, **VirtualBox**, or **Docker** as a driver).

Install one of the following:

**Hyper-V** (pre-installed on Windows 10/11 Pro or Enterprise).

**Docker Desktop** (if you want to use Docker as the driver).

## Step 2: Download Minikube

Open a PowerShell or Command Prompt with administrator privileges.

Download the latest Minikube executable using this command:

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-installer.exe
```

Install Minikube by running the installer:

```
.\minikube-installer.exe
```

## Step 3: Add Minikube to PATH

If Minikube is not automatically added to your PATH during installation:

Open **System Properties** → **Environment Variables**.

Add the directory where Minikube is installed (e.g., `C:\Program Files\Minikube`) to your PATH variable.

## Step 4: Start Minikube

Open a terminal (PowerShell or CMD). Do the following commands

Start Minikube with a specified driver (e.g., Hyper-V, Docker, or VirtualBox). For example:

**Hyper-V:**

```
minikube start --driver=hyperv
```

**Docker:**

```
minikube start --driver=docker
```

Verify Minikube is running:

```
minikube status
```

## Step 5: Interact with Minikube

kubectl is a command-line tool used in Kubernetes to interact with and manage Kubernetes clusters.

Once Minikube is running:

Use `kubectl` to interact with the cluster.

Install `kubectl` if not already installed:

```
minikube kubectl -- get pods -A
```

Or download it separately from the [official Kubernetes site](https://kubernetes.io/docs/tasks/tools/install-kubectl/).

Open the Minikube dashboard (optional):

```
minikube dashboard
```

Optional: Check Your Installation

**Run the following to verify the installation:**

Minikube version

```
kubectl version --client
```

## Troubleshooting

**If Minikube fails to start:**

Ensure your hypervisor (Hyper-V/Docker/VirtualBox) is installed and running.

Check the Minikube logs:

```
minikube logs
```

**Updating Minikube:**

```
minikube update-check
```

```
minikube update
```

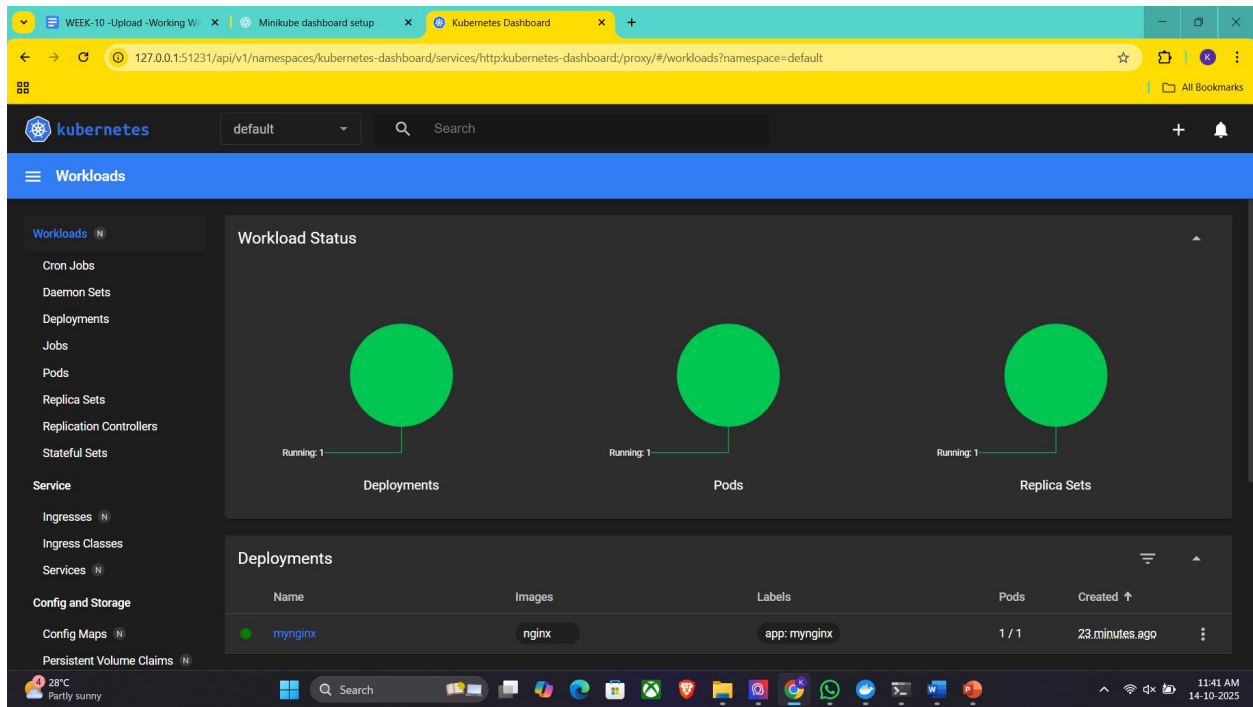
## Minikube Automation Steps

### Step 1: Start Minikube Cluster

Open your terminal and run the command:

```
minikube start
```





## Step 2: Create and Manage Deployment

Create an application in Kubernetes:

Command:

```
kubectl create deployment mynginx --image=nginx
```

if already created then

```
kubectl set image deployment/myngnix nginx=nginx:latest
```

Verify the deployment using: Kubernetes responds by showing you a list that includes the names of your deployment groups

```
kubectl get deployments
```

```
C:\Users\Asus>kubectl get deployments
NAME          READY    UP-TO-DATE    AVAILABLE    AGE
mynginx       1/1      1             1            32m

C:\Users\Asus>|
```

Ensure mynginx appears in the output.

Check the following commands:

kubectl get pods

kubectl describe pods

```
C:\Users\Asus>kubectl get pods
NAME          READY    STATUS    RESTARTS    AGE
mynginx-645865c456-87dfg  1/1      Running   0           32m

C:\Users\Asus>kubectl describe pods
Name:          mynginx-645865c456-87dfg
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 14 Oct 2025 11:18:05 +0530
Labels:        app=mynginx
               pod-template-hash=645865c456
Annotations:   <none>
Status:        Running
IP:            10.244.0.9
IPs:           IP: 10.244.0.9
               Controlled By: ReplicaSet/mynginx-645865c456
Containers:
  nginx:
    Container ID:  docker://7f27a5a82d0dc84202b8899d078fcb870a2bdc4ef4cc2c8815d50452bc821718
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:3b7732505933ca591ce4a6d860cb713ad96a3176b82f7979a8dfa9973486a0d6
    Port:          <none>
    Host Port:     <none>
    State:         Running
      Started:     Tue, 14 Oct 2025 11:22:52 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
```

## Expose Deployment as a Service:

Command:

```
kubectl expose deployment mynginx --type=NodePort --port=80 --
target-port=80
```

```
C:\Users\Asus>kubectl expose deployment mynginx --type=NodePort --port=80 --target-port=80
service/mynginx exposed

C:\Users\Asus>|
```

### Step 3: Scale the Deployment

Command: Scales the Nginx deployment to 4 replicas (pods).

```
kubectl scale deployment mynginx --replicas=4
```

```
kubectl get service myngnix
```

```
C:\Users\Asus>kubectl scale deployment mynginx --replicas=4
deployment.apps/mynginx scaled

C:\Users\Asus>kubectl get service myngnix
Error from server (NotFound): services "myngnix" not found

C:\Users\Asus>|
```

### Step 4: Access the Nginx App

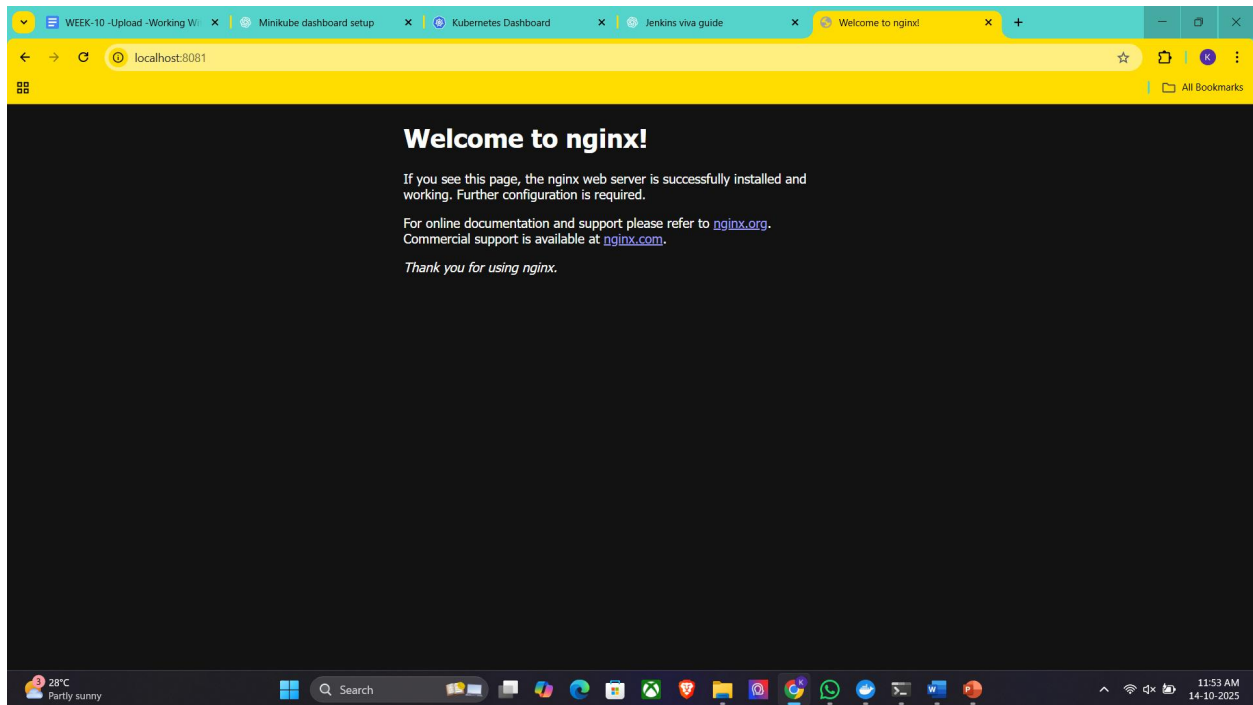
Using Port Forwarding:

Command:

```
kubectl port-forward svc/mynginx 8081:80
```

```
C:\Users\Asus>kubectl port-forward svc/mynginx 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
|
```

Access the app via <http://localhost:8081>.



If Error, use this option, **Using Minikube Tunnel:**

Start the tunnel:

```
minikube tunnel
```

Retrieve the service URL:

```
minikube service mynginx --url
```

```
C:\Users\Asus>minikube tunnel
* Tunnel successfully started

* NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

C:\Users\Asus>minikube service mynginx --url
http://127.0.0.1:57079
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

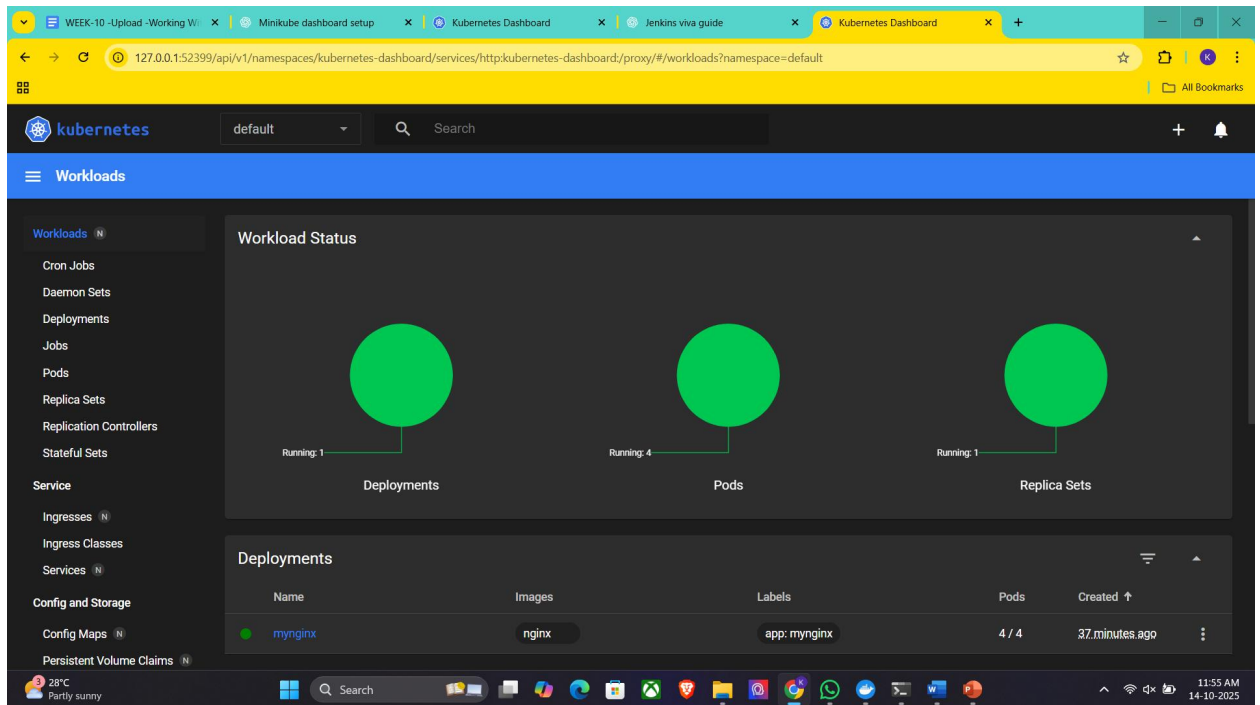
Open the provided URL in your browser.

Open the kubernetes dashboard

Open the minikube dashboard

Minikube dashboard





## Step 5: Stop and Clean Up

### Stop Nginx Deployment:

Commands:

```
kubectl delete deployment mynginx  
kubectl delete service mynginx
```

```
C:\Users\Asus>kubectl delete deployment mynginx  
deployment.apps "mynginx" deleted  
  
C:\Users\Asus>kubectl delete service mynginx  
service "mynginx" deleted  
  
C:\Users\Asus>
```

### Stop Minikube (Optional):

Command:

```
minikube stop
```

### Delete Minikube Cluster (Optional):

Command:minikube delete

## Nagios Automation Steps

### Step 1: Pull the Nagios Docker Image

Open a terminal and run:

```
docker pull jasonrivers/nagios:latest
```

```
C:\Users\Asus>docker pull jasonrivers/nagios:latest
latest: Pulling from jasonrivers/nagios
ff65ddf9395b: Pull complete
785b9873bdf4: Pull complete
0ef9446ba5cc: Pull complete
53aff88babc4: Pull complete
d72f92e29533: Pull complete
706ed7d4ce0a: Pull complete
d3245570f968: Pull complete
e58e184b986a: Pull complete
eeb77e6dde3e: Pull complete
0bd0f5795eeb: Pull complete
71bfb306f8cb: Pull complete
738fc7520889: Pull complete
fe8a6b2cf4e3: Pull complete
e6f8fab512d1: Pull complete
15f36d0b0439: Pull complete
a2fc4187e3b4: Pull complete
3d5785144815: Pull complete
566cdc02555d: Pull complete
c700be87d617: Pull complete
4f4fb700ef54: Pull complete
b69c76bd2b6b: Pull complete
d5aa2a3a6539: Pull complete
8fb30af17153: Pull complete
9ffe54c5c139: Pull complete
279b28aefa10: Pull complete
a900dfcceb38: Pull complete
9a90645e352c: Pull complete
8e911c59da28: Pull complete
c219d58cc3f9: Pull complete
b0e280e9aa8c: Pull complete
8c389e58e867: Pull complete
Digest: sha256:2a7c2b20d118baf92b47b69a3901e68dd7664617801b94e560bc4d6564d6ae54
Status: Downloaded newer image for jasonrivers/nagios:latest
docker.io/jasonrivers/nagios:latest
```

### Step 2: Run Nagios

Command:

```
docker run --name nagiosdemo -p 8888:80 jasonrivers/nagios:latest
```

```
C:\Users\Asus>docker run --name nagiosdemo -p 8888:80 jasonrivers/nagios:latest
Adding password for user nagiosadmin
shown: warning: '.' should be ':' 'nagios.nagios'
Started runsvdir, PID is 13
checking permissions for nagios & nagiosgraph
rsyslogd: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="20" x-info="https://www.rsyslog.com"] start

Nagios Core 4.5.7
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2024-10-24
License: GPL

Website: https://www.nagios.org
Nagios 4.5.7 starting... (PID=25)
Local time is Tue Oct 14 06:40:32 UTC 2025
wproc: Successfully registered manager as @wproc with query handler
nagios: Nagios 4.5.7 starting... (PID=25)
nagios: Local time is Tue Oct 14 06:40:32 UTC 2025
nagios: LOG VERSION: 2.0
nagios: qh: Socket '/opt/nagios/var/rw/nagios.qh' successfully initialized
nagios: qh: core query handler registered
nagios: qh: echo service query handler registered
nagios: qh: help for the query handler registered
nagios: wproc: Successfully registered manager as @wproc with query handler
wproc: Registry request: name=Core Worker 42;pid=42
nagios: wproc: Registry request: name=Core Worker 42;pid=42
wproc: Registry request: name=Core Worker 43;pid=43
nagios: wproc: Registry request: name=Core Worker 43;pid=43
wproc: Registry request: name=Core Worker 44;pid=44
nagios: wproc: Registry request: name=Core Worker 44;pid=44
wproc: Registry request: name=Core Worker 45;pid=45
nagios: wproc: Registry request: name=Core Worker 45;pid=45
wproc: Registry request: name=Core Worker 47;pid=47
nagios: wproc: Registry request: name=Core Worker 47;pid=47
wproc: Registry request: name=Core Worker 46;pid=46
```

### Step 3: Access Nagios Dashboard

Open your browser and navigate to:

`http://localhost:8888`

#### Login Credentials:

Username: nagiosadmin

Password: nagios

Once logged in, explore:

Hosts: View systems being monitored.

Services: Check tasks being monitored (e.g., CPU usage).

Alerts: Access recent notifications.

WEEK-10 -Upload -WorkinMinikube dashboard setupKubernetes DashboardJenkins viva guideKubernetes Dashboardlocalhost:8888/tree

localhost:8888/tree

Sign in

http://localhost:8888

Username

nagiosadmin

Password

\*\*\*\*\*

Sign in

Cancel

28°C  
Partly sunny

Search

WEEK-10 -Upload -WorkinMinikube dashboard setupKubernetes DashboardJenkins viva guideKubernetes DashboardNagios: localhost

localhost:8888

All Bookmarks

Home

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Host Groups

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Service Groups

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Network Outages

Quick Search:

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History

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Histogram

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Event Log

System

Comments

Downtime

Process Info

Performance Info

Notifications

Last Updated: Tue Oct 14 06:44:10 UTC 2025

Nagios® Core™ 4.5.7 - www.nagios.org

Logged in as nagiosadmin

View Status Detail For All Hosts

View History For All Hosts

All Hosts and Services

Latest Archive

Log File Navigation

Tue Oct 14 00:00:00 UTC 2025

to

Present..

File: /opt/nagios/var/nagios.log

Host Service Type Time Contact Notification Command Information

No notifications have been recorded in the current log file

Notification detail level for all hosts:

All notifications

Older Entries First:

Update

NIFTY

-0.31%

Search

12:14 PM  
14-10-2025

WEEK-10 -Upload -Workin

Minikube dashboard setup

Kubernetes Dashboard

Jenkins viva guide

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History

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Histogram

Notifications

Event Log

System

Comments

Downtime

Process Info

Performance Info

Current Network Status

Last Updated: Tue Oct 14 06:43:25 UTC 2025

Updated every 90 seconds

Nagios® Core™ 4.5.7 - www.nagios.org

Logged in as nagiosadmin

Host Status Totals

Up Down Unreachable Pending

1 0 0 0

All Problems All Types

0 1

Service Status Totals

Ok Warning Unknown Critical Pending

2 1 0 0 4

All Problems All Types

1 7

View History For all hosts

View Notifications For All Hosts

View Host Status Detail For All Hosts

Service Status Details For All Hosts

Limit Results: 100

| Host      | Service         | Status  | Last Check          | Duration      | Attempt | Status Information                                                                |
|-----------|-----------------|---------|---------------------|---------------|---------|-----------------------------------------------------------------------------------|
| localhost | Current Load    | OK      | 10-14-2025 06:41:14 | 0d 0h 2m 53s+ | 1/4     | OK - load average: 0.56, 0.60, 0.45                                               |
|           | Current Users   | OK      | 10-14-2025 06:41:57 | 0d 0h 2m 53s+ | 1/4     | USERS OK - 0 users currently logged in                                            |
|           | HTTP            | WARNING | 10-14-2025 06:42:40 | 0d 0h 0m 45s  | 1/4     | HTTP WARNING: HTTP/1.1 401 Unauthorized - 695 bytes in 0.000 second response time |
|           | PING            | PENDING | N/A                 | 0d 0h 2m 53s+ | 1/4     | Service check scheduled for Tue Oct 14 06:43:23 UTC 2025                          |
|           | Root Partition  | PENDING | N/A                 | 0d 0h 2m 53s+ | 1/4     | Service check scheduled for Tue Oct 14 06:44:06 UTC 2025                          |
|           | Swap Usage      | PENDING | N/A                 | 0d 0h 2m 53s+ | 1/4     | Service check scheduled for Tue Oct 14 06:44:49 UTC 2025                          |
|           | Total Processes | PENDING | N/A                 | 0d 0h 2m 53s+ | 1/4     | Service check scheduled for Tue Oct 14 06:45:32 UTC 2025                          |

Results 1 - 7 of 7 Matching Services

Page Tour

NIFTY

-0.31%

Search

12:13 PM

14-10-2025

The top screenshot displays the Nagios web interface at localhost:8888. The left sidebar contains a navigation menu with sections: Home, Documentation, Current Status, Tactical Overview, Map, Hosts, Services, Host Groups, Service Groups, Problems, Reports, Availability, Trends, Alerts, Notifications, System, and Comments. The main content area shows 'Current Network Status' (last updated Tue Oct 14 06:43:13 UTC 2025), 'Host Status Totals' (Up: 1, Down: 0, Unreachable: 0, Pending: 0), and 'Service Status Totals' (Ok: 2, Warning: 1, Unknown: 0, Critical: 0, Pending: 4). Below these is a table titled 'Host Status Details For All Host Groups' with columns: Host, Status, Last Check, Duration, and Status Information. The table lists 'localhost' with status 'UP', last check '10-14-2025 06:42:40', duration '0d 0h 2m 37s', and status information 'PING OK - Packet loss = 0%, RTA = 0.04 ms'. The bottom screenshot shows the Nagios landing page (Version 4.5.7, October 24, 2024) with a 'Meet Nagios Core Services Platform' section, a 'Check for updates' button, and links for 'Learn More' and 'Newsletter Sign-Up'.

## Step 4: Monitoring Host Details

Navigate to the Host Information Page:

Select a host from the **Hosts** menu.

**Key Details:**

Host Status: Indicates if the system is UP or DOWN.

Metrics: View CPU usage, memory status, and network activity.

Actions: Reschedule checks, disable notifications, or schedule downtime.

The screenshot shows the Nagios localhost interface. The top navigation bar is yellow. The left sidebar contains links for Home, Documentation, Current Status, Tactical Overview, Map, Hosts, Services, Host Groups, Summary, Grid, Service Groups, Summary, Grid, Problems, Services (Unhandled), Hosts (Unhandled), Network Outages, Quick Search, Reports, Availability, Trends, Alerts, History, Summary, Histogram, Notifications, Event Log, System, Comments, Downtime, Process Info, and Performance Info. The main content area displays the 'Current Network Status' (Last Updated: Tue Oct 14 06:45:47 UTC 2025, Updated every 90 seconds, Nagios® Core™ 4.5.7 - www.nagios.org, Logged in as nagiosadmin). Below this are 'Host Status Totals' (Up: 1, Down: 0, Unreachable: 0, Pending: 0) and 'Service Status Totals' (Ok: 6, Warning: 1, Unknown: 0, Critical: 0, Pending: 0). The 'Host Status Details For All Host Groups' table shows one host, 'localhost', with a status of 'UP', last check of '10-14-2025 06:42:40', duration of '0d 0h 5m 11s', and status information 'PING OK - Packet loss = 0%, RTA = 0.04 ms'. The bottom status bar shows 'AFG - BAN In 6 hours' and the system clock '12:15 PM 14-10-2025'.

| Host      | Status | Last Check          | Duration     | Status Information                        |
|-----------|--------|---------------------|--------------|-------------------------------------------|
| localhost | UP     | 10-14-2025 06:42:40 | 0d 0h 5m 11s | PING OK - Packet loss = 0%, RTA = 0.04 ms |

The screenshot shows the Nagios localhost interface. The top navigation bar is yellow. The left sidebar contains links for Home, Documentation, Current Status, Tactical Overview, Map, Hosts, Services, Host Groups, Summary, Grid, Service Groups, Summary, Grid, Problems, Services (Unhandled), Hosts (Unhandled), Network Outages, Quick Search, Reports, Availability, Trends, Alerts, History, Summary, Histogram, Notifications, Event Log, System, Comments, Downtime, Process Info, and Performance Info. The main content area displays the 'Current Network Status' (Last Updated: Tue Oct 14 06:45:39 UTC 2025, Updated every 90 seconds, Nagios® Core™ 4.5.7 - www.nagios.org, Logged in as nagiosadmin). Below this are 'Host Status Totals' (Up: 1, Down: 0, Unreachable: 0, Pending: 0) and 'Service Status Totals' (Ok: 6, Warning: 1, Unknown: 0, Critical: 0, Pending: 0). The 'Service Status Details For All Host Groups' table shows seven services for the host 'localhost'. The 'HTTP' service is in a 'WARNING' state. The bottom status bar shows 'Air: Satisfactory Tomorrow' and the system clock '12:16 PM 14-10-2025'.

| Host      | Service         | Status  | Last Check          | Duration     | Attempt | Status Information                                                                |
|-----------|-----------------|---------|---------------------|--------------|---------|-----------------------------------------------------------------------------------|
| localhost | Current Load    | OK      | 10-14-2025 06:46:14 | 0d 0h 6m 25s | 1/4     | OK - load average: 0.07 0.26 0.35                                                 |
|           | Current Users   | OK      | 10-14-2025 06:41:57 | 0d 0h 6m 7s+ | 1/4     | USERS OK - 0 users currently logged in                                            |
|           | HTTP            | WARNING | 10-14-2025 06:45:40 | 0d 0h 0m 59s | 4/4     | HTTP WARNING: HTTP/1.1 401 Unauthorized - 695 bytes in 0.000 second response time |
|           | PING            | OK      | 10-14-2025 06:43:23 | 0d 0h 6m 7s+ | 1/4     | PING OK - Packet loss = 0%, RTA = 0.05 ms                                         |
|           | Root Partition  | OK      | 10-14-2025 06:44:06 | 0d 0h 6m 7s+ | 1/4     | DISK OK - free space: / 969708 MB (99.09% inode=100%)                             |
|           | Swap Usage      | OK      | 10-14-2025 06:44:49 | 0d 0h 6m 7s+ | 1/4     | SWAP OK - 100% free (2048 MB out of 2048 MB)                                      |
|           | Total Processes | OK      | 10-14-2025 06:45:32 | 0d 0h 6m 7s+ | 1/4     | PROCS OK: 41 processes with STATE = RSZDT                                         |

## Step 5: Stop and Remove Nagios

### Stop the Container:

Command:

```
docker stop nagiosdemo
```

```
C:\Users\Asus>docker stop nagiosdemo
nagiosdemo
```

### Delete the Container:

Command:

```
docker rm nagiosdemo
```

```
C:\Users\Asus>docker rm nagiosdemo
nagiosdemo
```

### Remove the Image (Optional):

List images:

```
docker images
```

```
C:\Users\Asus>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
gcr.io/k8s-minikube/kicbase   latest             f5b47042e2a5       8 weeks ago        137MB
redisnew             latest             9d1fe3a9a889       8 weeks ago        137MB
redis                mylatest           9d1fe3a9a889       8 weeks ago        137MB
ubuntuos             latest             ba0725824e67       5 months ago       347MB
<none>               <none>             0dea92c08728       7 months ago       347MB
<none>               <none>             6a51c5ed6fdf       7 months ago       347MB
<none>               <none>             86d36e5d4bae       7 months ago       347MB
<none>               <none>             75be11f731f5       7 months ago       347MB
linux1               latest             0d3955e89158       7 months ago       331MB
<none>               <none>             57ff21045edb       7 months ago       331MB
<none>               <none>             85107a5f3881       7 months ago       341MB
ubuntu               latest             a04dc4851cbc       8 months ago       78.1MB
jasonrivers/nagios   latest             2ab73fb5d162       11 months ago      960MB
docker/desktop-kubernetes  v1.30.5            a60dd8dae2ad       12 months ago      435MB
registry.k8s.io/kube-apiserver  v1.30.5            e9ad5c076a8        13 months ago      117MB
registry.k8s.io/kube-scheduler  v1.30.5            25903461a65c       13 months ago      62MB
registry.k8s.io/kube-controller-manager  v1.30.5            38406042cf08       13 months ago      111MB
registry.k8s.io/kube-proxy      v1.30.5            71161e05b9bb       13 months ago      84.7MB
registry.k8s.io/coredns/coredns  v1.11.3            c69fa2e9cbf5       14 months ago      61.8MB
registry.k8s.io/etcd            3.5.15-0           2e96e5913fc0       14 months ago      148MB
registry.k8s.io/etcd            3.5.12-0           3861cfc7c04        20 months ago      149MB
docker/desktop-vpnkit-controller  dc331cb22850be0cdd97c84a9cfecaf44a1afb6e  3.9                556098075b3d       2 years ago        36.2MB
registry.k8s.io/pause          3.9                e6f181688397       3 years ago         744KB
docker/desktop-storage-provisioner  v2.0               99f89471f470       4 years ago         41.9MB
C:\Users\Asus>
```

Delete the Nagios image:

```
docker rmjasonrivers/nagios:latest
```



Observe the docker containers in DockerHub, we can see the latest Nagios  
Installed running on port:8888

```
C:\Users\Asus>docker rmi jasonrivers/nagios:latest
Untagged: jasonrivers/nagios:latest
Untagged: jasonrivers/nagios@sha256:2a7c2b20d118baf92b47b69a3901e68dd7664617801b94e560bc4d6564d6ae54
Deleted: sha256:2ab73fb5d162ad0be4ef495fb62e17772120b32721a6c3eb9091fcc800d2909b
Deleted: sha256:631ab139f760d68457a4186da4a60f5e5f11e65aa3b26864b5df237ef75b1311
Deleted: sha256:b10e2a7f4de8bcf15a69232335422c9a736c4a4a93ca288592a2fb53a2de6f45
Deleted: sha256:3a5a34bf7e5ab76f1a3054c5854a9bc7227fd5a0dec73d4a4673456f69cec753
Deleted: sha256:0603355dfe22b531eff49627391c7a8bb944163a735d9142cfcf02529e3c8fffd
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```

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## Steps for AWS-free trier account creation

open <https://aws.amazon.com>

click on create AWS account

Provide email id and name details for AWS account creation then a screen will appear as below

Enter verification code received in your mail

Provide your contact details and agree Terms and conditions, then click on create

Provide billing details, click on verify and continue

Amazon charges 2 rs and it will be credit back into your account once verification is over

Once payment success a screen appear as below, provide your working contact number after selecting country, and click on send SMS after entering the details

Provide verification code received in your mobile

Now select basic plan and click on complete sign up

Now a screen appear as follows then click on AWS management console

You will navigating to the aws page ..where you need to Select role as Academic/Researcher and interest as DevOps and click on submit

Sign into AWS console as Root user

A console screen appears aws will appear

**Conclusion:** In this week we learnt how to deploy pods in minikube and how to monitor our local system using Nagios tool and will know how to create the aws free trier account.